

Networking Systems and Social Computing and Principles of Cybersecurity

Year 11 Preliminary Course Enterprise Computing Task 2



HURLSTONE
AGRICULTURAL HIGH SCHOOL

Name: William Lam

Class: 11ECOMA

Teacher: Ms Hidalgo

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Logbook

Logbook				
Date	Objective	Activity Completed	Challenges	To be completed
17/06/25	Read Notification	Notification was read and understood. Planning was started with the creation of the folio.	The plan was quite simple and not as detailed as I would've liked.	Begin work on Cisco
23/07/25	Cloud computing exercise	Completed an activity on cloud computing in class Upon arriving home I created the network diagram	Did not work on Cisco Creating the network diagram took some time, as my planning was not thorough enough	Begin work on Cisco
25/07/25	Begin work on Cisco	Started the creation of the IT and Accounting Department	Navigating the UI of Cisco was quite difficult	Continue Cisco Network
29/07/25	Continue Cisco network	The accounting department was finished and the Sales department was started and finished. I also did a cybersecurity exercise. Upon arriving home, I created and configured all departments, as well as the ISP server	When creating the Admin Department, I struggled to connect the Iot Devices (Smoke detector and webcam) to the network, requiring further research. Also I managed to crash Cisco and lost about an hour's work. I had multiple attempts trying to setup the IoT devices, eventually realising the method was to use a gateway instead of directly connecting it to a switch.	Create firewall for network

			The ISP server was extremely difficult to work with, taking me close to an hour to figure out how to use.	
30/07/25	Create firewall	Continued the cybersecurity exercise from last lesson, as well as clarified the naming conventions. Created a basic firewall on the ISP server, allowing IP connections. Upon arriving home I set up WPA2-PSK authentication for all the wireless devices. I realised that I can only use endpoints in physical mode, hence tidying up the cabling in the physical layout. To secure the ethernet ports, the mac addresses of the current devices connected were set, and violations were in place to stop new mac addresses from joining. If a laptop manages to connect to the wifi and isn't verified, the wifi network will shut down. Any unused ports on the switches have been turned off as well.	I am unsure if the firewall on the ISP server is adequate enough. I would also like to make the logical component of my network more visually pleasing by adding in bends. I attempted to use AAA but could not work it out. I believe I have to use CLI which the wireless router does not have, meaning there is no security for the devices connected physically.	Rewire physical mode, as setting up the firewall moved a lot of cabling. Consider WPA2 Enterprise

31/07/25	Continue firewall	I swapped my router from WRT300N to the Home Router. I did this because the Home Router seemed better suited as it has both 2.4ghz and 5ghz, as well as easily accessible guest networks. In physical mode, I moved the departments around to better suit the marking criteria. I upgraded the network to gigabit connection using fiber, starting with the ISP. I will have to check if using fiber is allowed. I also configured the IoT devices, setting up a registration server.	When configuring the IoT devices, I had it originally working then it just stopped and wouldn't connect to the network, which was frustrating. I was unable to find the cause of it originally.	Consider mesh topology and reattempt AAA
01/08/25	Network Logical change	I changed the logical view of the network from a star topology to a tree topology. I decided to use tree topology over mesh topology due to ease of use. I updated the naming of the devices to a more consistent naming convention. I began documenting the network into my project folio. Did a significant portion of the network overview, documenting the IPs and MAC addresses of each device.	Formatting the folio took some time as my original plan did not align with my current plan. AAA was not reattempted as there was no need for it due to it not being in the marking criteria.	Continue on folio work Consider changing naming convention

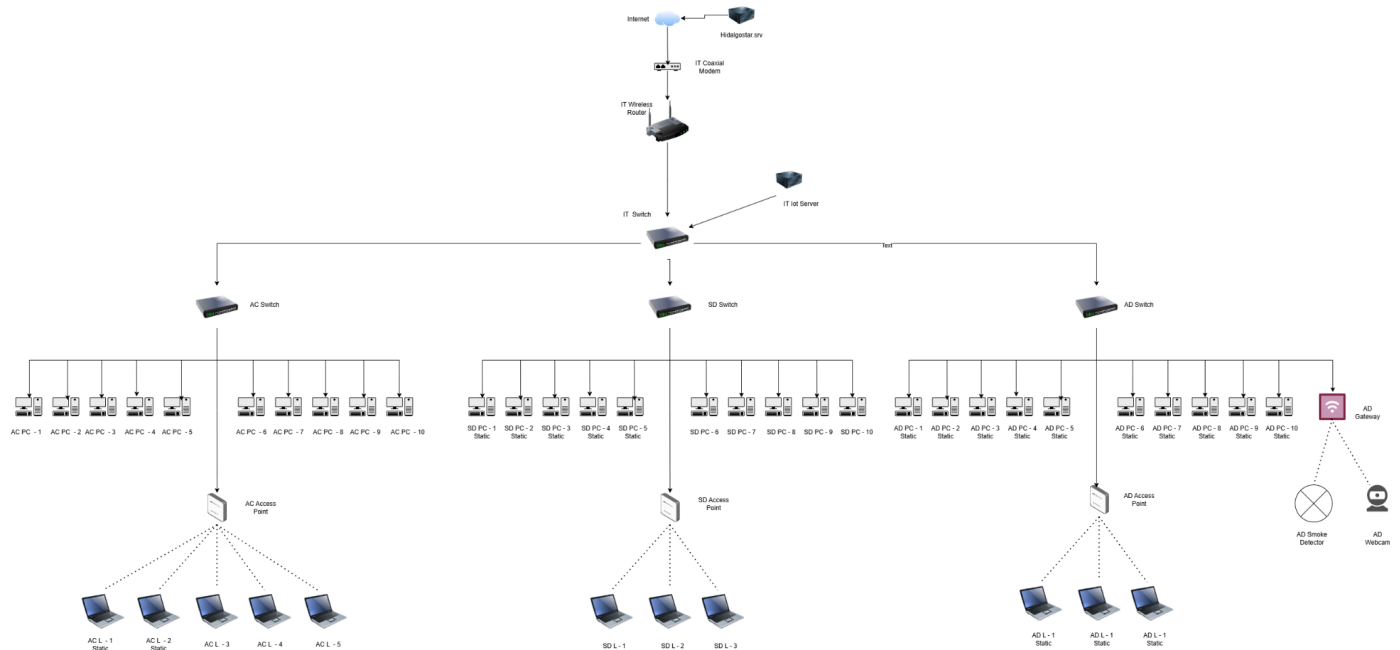
		For fun, since I had already configured the entire network, I added 2 html servers, using html files I created in year 9 for previous assignments. They can be accessed by going to https://www.williamlam.net and https://www.adampham.com in the web browser on a pc/laptop (it will take a long time to load due to images).		
02/08/25	Continue Folio Work	I fixed up a bit of information in the tables; I had entered it incorrectly. I tweaked a few of the DNS servers.	I wanted to swap my entire network to fiber (from copper straight through), however there were two main things stopping me: the notification stated to use copper straight through cables, and the switches which supported fiber connections capped out at a maximum of 4, meaning I would need switches connected to other switches to provide enough gigabit ports for all devices.	Start on cybersecurity section of folio
04/08/25	Risk Matrix	I created a risk matrix with 5 levels of impact and likelihood, as well as created a table that will be filled in with the risks and their respective scores.	I could not start filling in the table due to not having completed the threat analysis.	Begin threat analysis

05/08/25	Threat Analysis	Before beginning my threat analysis, I realised that you did not need fiber to have a 1000mbps connection. I just had to swap to GigabitEthernet for all my devices. I then swapped my switches from the 2960 to the 3650, which had both fiber and gigabit ethernet connections. I had to reconfigure just about every device on my network, and managed to have a 1000mbps for all my computers. I connected my access points via fiber. In the threat analysis, I created an outline of the issues I want to discuss, with basic preliminary information, which I will expand on later.	Did not get through a lot of the threat analysis as I was upgrading the network to gigabit connections.	Continue Threat analysis
08/08/25	Threat Analysis	I put in a vast majority of my information for the threat analysis	I feel like I may have too much information	Cut down and continue the cybersecurity section.
22/08/25	Cutting down threat analysis	I cut down the threat analysis sizeable and completed it. I also completed the risk matrix.	Scoring the incidents was difficult.	Start mitigation plan.

26/08/25	Completed mitigation plan	I completed the mitigation plan, meaning the folio is completed.	I spent most of my time dedicated to working on this assignment helping others, meaning I stayed up quite late doing my mitigation plan.	Video
27/05/25	Video was shot	It is currently 12:55am. I have finished recording my videos and will edit them Update: Time is now 1:50am. I have finished editing all the videos and trimmed down significantly to 14 minutes and size has been cut down to 300mb. Can be found here: https://drive.google.com/file/d/1Ytf3OHRBGiiTcDGVZfZ_wSBtn8Szyihw/view?usp=sharing Update (6:05pm): I have cut down to 6:46 which will be my final video	My video is currently sitting around 22 minutes, I will have to check to see if I am allowed to upload a 1.63 gigabyte video file. The video is still insanelly long.	Review and Submit
28/05/25	Submission	Folio, cisco and video were all reviewed. Submitted.	NIL	NIL

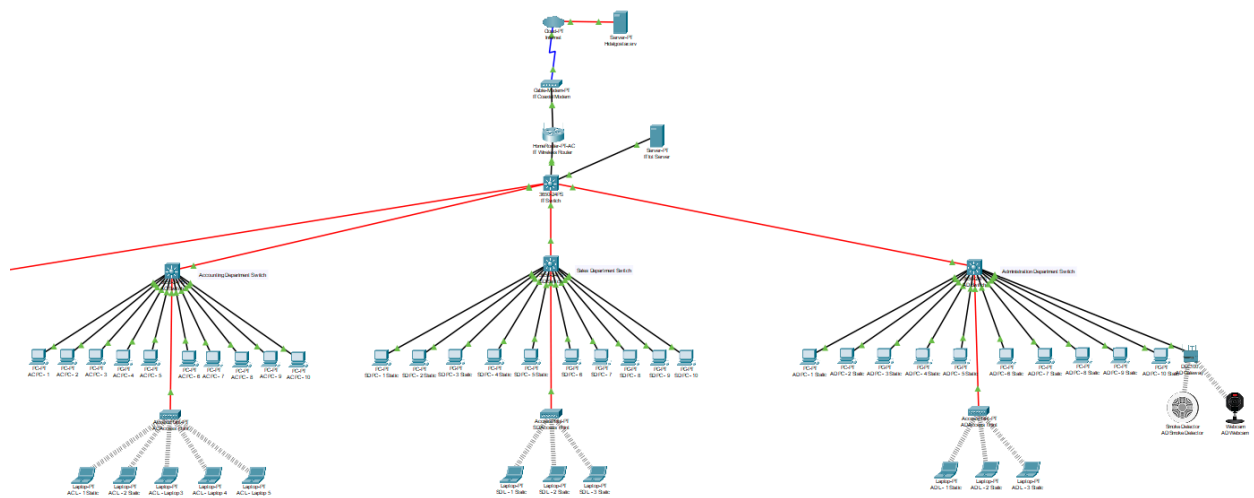
Network Overview

Network Diagram



Higher Resolution Image can be found here:

<https://drive.google.com/file/d/1TXdROAx9v1iCjqHEPczW4pAD4RdZrG5J/view?usp=sharing>



Security Overview

To secure the wireless connections, I used WPA2 Personal and WPA2-PSK, which requires a password.

For the switches, I turned off the GigabitEthernet ports which were not in use (GigabitEthernet1/0/11 - GigabitEthernet1/0/23 on AC and SD, GigabitEthernet1/0/12 - GigabitEthernet0/23 on AD, as well as GigabitEthernet1/1/2 - GigabitEthernet1/1/4 on all). This prevents PCs from connecting to an unused port. On the GigabitEthernet0/2 ports, I

set a **maximum number of connections** corresponding to the number of laptops required from the notification. If that number is breached, the access point will close and be inaccessible until it is manually turned back on or network is reset.

Additionally, all the mac addresses of the PCs and Laptops have been made **'sticky'**, and violation set to **shutdown**. This means that if a PC/Laptop were to somehow make it past all the previous barriers, it would be blocked from connecting to the network as its MAC address is not on the list of the switches. One of the flaws of this was the security of a port in use by a PC. If a new pc with a new MAC address connects to that port, the network remains open, meaning the system has a vulnerability there.

```
interface FastEthernet0/24
  shutdown
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
  switchport mode access
  switchport port-security
  switchport port-security maximum 3
  switchport port-security mac-address sticky
  switchport port-security mac-address sticky 0001.42AA.9E1B
  switchport port-security mac-address sticky 0090.0CCB.9533
  switchport port-security mac-address sticky 00E0.F7A3.69DB
!
interface FastEthernet0/10
  switchport mode access
  switchport port-security
  switchport port-security mac-address sticky
  switchport port-security mac-address sticky 0003.E414.BD10
!
```

Wireless Networks

Wireless Networks			
Department	SSID	Password	Type of Security
IT	IT Department 2.4ghz	Gd3hK9pJ1m	WPA2 Personal
IT	IT Department 5ghz	Cq5fR3kT8h	WPA2 Personal
IT	IT Guest 2.4GHZ		Disabled
IT	IT Guest 5GHZ		Disabled
Accounting	Accounting Department	Rp7sT2qX5b	WPA2-PSK
Sales	Sales Department	Za1oC8dN4f	WPA2-PSK

Administration	Administration Department	Ue9wM6yL0v	WPA2-PSK
Registration Server (IoT)			
Username		Password	
IotAdmin		Vz6eN4dP9u	
IT Wireless Router			
Username		Password	
admin		admin	

DNS Service

DNS			
Domain (what you type into the web browser)	Type	Address (what the site ends up taking you to)	Description
www.hidalgostar.com	A Record	210.158.211.125	Website where users can see and interact with the IoT devices.
hidalgostar.srv	CNAME	www.hidalgostar.com	
hidalgostar.com	CNAME	www.hidalgostar.com	
www.adampham.com	A Record	192.168.0.5	Website about Adam Pham
adam.net	CNAME	www.adampham.com	
adampham.com	CNAME	www.adampham.com	
www.williamlam.net	A Record	192.168.0.6	Website about William Lam
williamlam.net	CNAME	www.williamlam.net	

A Record: Maps a domain to an IPv4 Address

CNAME: Maps a domain to an already existing domain

When attempting to reach a domain, you **MUST** use **https://** before typing in the domain name, otherwise it will default to http://. This is done to better secure the domains and the network.

Devices

Naming Convention

E.g.

SD PC - 5 Static

The first two letters symbolise the department.

- IT = It Department
- AC = Accounting Department
- SD = Sales Department
- AD = Administration Department

It will then be followed by the type of device.

- Wireless Router
- Coaxial Modem
- PC (Personal Computer)
- L (Laptop)
- Switch
- Smoke Detector
- Webcam
- Gateway
- Access Point
- Iot Server

After the hyphen (if applicable), there will be a number if there are duplicates (e.g. the 10 PCs will have numbers 1-10).

Then it will be followed by 'Static' if the device has a Static IP (meaning it never changes). If it does not say 'Static', the device's IP is handled by the DHCP service (meaning it may not always be the same ip).

This naming convention applies to everything except the ISP, Internet and Gigabit Switch and its servers.

Devices

ISP: Hidalgostar.srv

Default Gateway: 192.168.0.1

DNS: 210.158.211.125

Devices					
ISP					
Internet					
Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
Hidalgostar.srv	210.158.211.125	00E0.F708.5026	Fiber	Internet	GigabitEthernet9
IT Coaxial Modem	NIL	NIL	Coaxial	Internet	Coaxial7

IT Department					
IT Wireless Router					
Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
IT Coaxial Modem	NIL	NIL	Copper Straight-Through	IT Wireless Router	Internet
IT Switch	NIL	00:02:16:3E:6A:A9	Copper Straight-Through	IT Wireless Router	GigabitEthernet0/1
IT Switch					
Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
IT Wireless Router	192.168.0.1	000D.BD05.1C01	Copper Straight-Through	IT Switch	GigabitEthernet1/0/1
AC Switch	NIL	00:0C:CF:39:3E:2E	Fiber	IT Switch	GigabitEthernet1/1/2
SD Switch	NIL	00:40:0B:EC:A5:E2	Fiber	IT Switch	GigabitEthernet1/1/3
AD Switch	NIL	00:07:EC:78:78:06	Fiber	IT Switch	GigabitEthernet1/1/4
IT Iot Server	192.168.0.7	0006.2AC2.D236	Copper Straight-Through	IT Switch	GigabitEthernet1/0/2
Gigabit Switch	NIL	00:D0:97:27:94:65	Fiber	IT Switch	GigabitEthernet1/1/1
The Gigabit Switch was not something mentioned on the notification, it links servers which contain previous HTML websites I have created. I used the gigabit switch as something to better further my skills in and understanding of Cisco Packet Tracer.					
Accounting Department					
AC Switch					
Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
IT Switch	NIL	00:02:16:3E:6A:A9	Fiber	AC Switch	GigabitEthernet1/1/4
AC PC - 1	DHCP	0004.9AA4.969A	Copper Straight-Through	AC Switch	GigabitEthernet1/0/1
AC PC - 2	DHCP	000C.CF6D.AB04	Copper Straight-Through	AC Switch	GigabitEthernet1/0/2

AC PC - 3	DHCP	000B.BE07.AB63	Copper Straight-Throug h	AC Switch	GigabitEthernet1 /0/3
AC PC - 4	DHCP	0001.960D.500D	Copper Straight-Throug h	AC Switch	GigabitEthernet1 /0/4
AC PC - 5	DHCP	0060.5CC2.EBA3	Copper Straight-Throug h	AC Switch	GigabitEthernet1 /0/5
AC PC - 6	DHCP	00E0.F7D2.46EA	Copper Straight-Throug h	AC Switch	GigabitEthernet1 /0/6
AC PC - 7	DHCP	000D.BD5E.6A01	Copper Straight-Throug h	AC Switch	GigabitEthernet1 /0/7
AC PC - 8	DHCP	0001.97EB.15B9	Copper Straight-Throug h	AC Switch	GigabitEthernet1 /0/8
AC PC - 9	DHCP	0002.172E.6B97	Copper Straight-Throug h	AC Switch	GigabitEthernet1 /0/9
AC PC - 10	DHCP	0007.EC03.C644	Copper Straight-Throug h	AC Switch	GigabitEthernet1 /0/10
AC Access Point	NIL	NIL	Fiber	AC Switch	GigabitEthernet1 /1/1

The reason why the AC Access Point has no IPv4 nor MAC address is because it is the basic AC on Cisco, meaning it has limited functionality and cannot show IPv4 or MAC.

AC Access Point

Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
AC Switch	NIL	00:0C:CF:39:3E:2E	Fiber	AC Access Point	Port 0
AC L - 1 Static	192.168.0 .201	00E0.F988.41D6	Wireless	AC Access Point	Port1
AC L - 2 Static	192.168.0 .200	0050.0FAD.26DE	Wireless	AC Access Point	Port1
AC L - 3 Static	DHCP	000A.410C.BC52	Wireless	AC Access Point	Port1

AC L - 4 Static	DHCP	0090.0C89.4857	Wireless	AC Access Point	Port1
AC L - 5 Static	DHCP	0001.4268.B666	Wireless	AC Access Point	Port1
Sales Department					
SD Switch					
Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
IT Switch	NIL	00:02:16:3E:6A:A9	Fiber	AC Switch	GigabitEthernet1 /1/4
SD PC - 1	192.168.0 .150	000C.8521.A461	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/1
SD PC - 2	192.168.0 .151	000A.F3AB.A58C	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/2
SD PC - 3	192.168.0 .152	0090.0CCA.7D79	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/3
SD PC - 4	192.168.0 .153	00D0.D344.9634	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/4
SD PC - 5	192.168.0 .154	000D.BD41.5114	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/5
SD PC - 6	DHCP	0001.9648.6667	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/6
SD PC - 7	DHCP	0060.2F30.D2EB	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/7
SD PC - 8	DHCP	0040.0B81.9BDA	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/8
SD PC - 9	DHCP	0030.F207.DA12	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/9
SD PC - 10	DHCP	0003.E43B.3AE7	Copper Straight-Thr ough	SD Switch	GigabitEthernet1 /0/10
AC Access	NIL	NIL	Fiber	SD	GigabitEthernet1

Point				Switch	/1/1
SD Access Point					
Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
SD Switch	NIL	00:40:0B:EC:A5:E2	Fiber	AC Access Point	Port 0
SD L - 1 Static	192.168.0.210	00E0.F7A3.69DB	Wireless	SD Access Point	Port1
SD L - 2 Static	192.168.0.211	0090.0CCB.9533	Wireless	SD Access Point	Port1
SD L - 3 Static	192.168.0.212	0001.42AA.9E1B	Wireless	SD Access Point	Port1
Administration Department					
AD Switch					
Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
IT Switch	NIL	00:02:16:3E:6A:A9	Fiber	AC Switch	GigabitEthernet1/1/4
AD PC - 1 Static	192.168.0.160	00D0.BC21.AD80	Copper Straight-Through	AD Switch	GigabitEthernet1/0/1
AD PC - 2 Static	192.168.0.161	0009.7C8B.999D	Copper Straight-Through	AD Switch	GigabitEthernet1/0/2
AD PC - 3 Static	192.168.0.162	0001.C7C5.9473	Copper Straight-Through	AD Switch	GigabitEthernet1/0/3
AD PC - 4 Static	192.168.0.163	0001.42D1.57C9	Copper Straight-Through	AD Switch	GigabitEthernet1/0/4
AD PC - 5 Static	192.168.0.164	000C.CFD8.06DB	Copper Straight-Through	AD Switch	GigabitEthernet1/0/5
AD PC - 6 Static	192.168.0.165	0060.7084.BE83	Copper Straight-Through	AD Switch	GigabitEthernet1/0/6
AD PC - 7 Static	192.168.0.166	0000.0C1B.E944	Copper Straight-Through	AD Switch	GigabitEthernet1/0/7

AD PC - 8 Static	192.168.0 .167	0001.6326.AB90	Copper Straight-Thr ough	AD Switch	GigabitEthernet1 /0/8
AD PC - 9 Static	192.168.0 .168	00E0.F940.398D	Copper Straight-Thr ough	AD Switch	GigabitEthernet1 /0/9
AD PC - 10 Static	192.168.0 .169	000C.CF16.4A89	Copper Straight-Thr ough	AD Switch	GigabitEthernet1 /0/10
AD Gateway	DHCP	0030.F2BE.E601	Copper Straight-Thr ough	AD Switch	GigabitEthernet1 /0/11
AC Access Point	NIL	NIL	Fiber	AD Switch	GigabitEthernet1 /1/1

AD Gateway

Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
AD Switch	NIL	00:07:EC:78:78:06	Copper Straight-Thr ough	AD Gateway	Internet
AD Smoke Detector	DHCP	0001.4347.81BA	Wireless	AD Gateway	Wireless
AD Webcam	DHCP	0001.64A5.B3A8	Wireless	AD Gateway	Wireless

AD Smoke Detector and AD Webcam are connected to a remote IoT server, 192.168.0.7

AD Access Point

Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
AD Switch	NIL	00:07:EC:78:78:06	Fiber	AC Access Point	Port 0
AD L - 1 Static	192.168.0 .220	0000.0C0A.B88B	Wireless	AD Access Point	Port1
AD L - 2 Static	192.168.0 .221	00E0.B016.97BE	Wireless	AD Access Point	Port1
AD L - 3 Static	192.168.0 .222	0001.4229.48C9	Wireless	AD Access Point	Port1

Gigabit Switch

Name	IPv4	MAC	Con. Type	Con. To	Port of Con.
------	------	-----	-----------	---------	--------------

IT Switch	NIL	00:02:16:3E:6A:A9	Fiber	AC Switch	GigabitEthernet1/1/4
adam.srv	192.168.0.5	0002.4AE7.DDAB	Fiber	Gigabit Switch	GigabitEthernet1/1/1
william.srv	192.168.0.6	0005.5ED1.EE50	Fiber	Gigabit Switch	GigabitEthernet1/1/2

The Gigabit Switch connects servers to the IT Wireless Router. These servers are HTML websites I had previously created in IST in Year 9 and 10. I found it interesting to be able to link my old work into Cisco Packet Tracer, hence its inclusion in my assignment.

Useful Commands (Use in CLI of respective switch) to check information:

Steps:

First press ENTER

Then type in enable

Press RETURN to get started!

Switch>enable|

Then type one of the following commands:

show running-config

- Hold/Press ENTER until all information is displayed on the console

```
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
switchport mode access
switchport port-security
switchport port-security maximum 5
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.4268.B666
switchport port-security mac-address sticky 000A.410C.BC52
switchport port-security mac-address sticky 0050.0FAD.26DE
switchport port-security mac-address sticky 0090.0C89.4857
switchport port-security mac-address sticky 00E0.F988.41D6
!
interface Vlan1
no ip address
shutdown
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
end
```

- (go up to red at minimum)
- Can be used to check the mac addresses of the PCs connected to the switch (shown in green)

```

interface FastEthernet0/8
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00E0.F9E6.76A8
- !

```

- To make sure the mac addresses are correct for the laptops and IoT devices, it is best to go into their respective config>wireless0 where it will then say the MAC address (shown in green)



show version

```

cisco WS-C2960-24TT-L (PowerPC405) processor (revision B0) with 65536K bytes of memory.
Processor board ID FOC1010X104
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.

64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address      : 00:60:2F:C4:E2:41
Motherboard assembly number    : 73-10390-U03
Power supply part number       : 341-0097-02
Motherboard serial number      : FOC10093R12
Power supply serial number     : AZS1007032H
Model revision number          : B0
Motherboard revision number    : B0
Model number                   : WS-C2960-24TT-L
System serial number           : FOC1010X104
Top Assembly Part Number       : 800-27221-02
Top Assembly Revision Number   : A0
Version ID                     : V02
CLEI Code Number               : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version  SW Image
-----
*    1 26    WS-C2960-24TT-L    15.0(2)SE4    C2960-LANBASEK9-M

- Configuration register is 0xF

```

- Used to find mac addresses of the switches (shown in green)
- This information is also available when first opening the CLI

Cybersecurity

Threat analysis

Technical Vulnerabilities

System and Network Configuration Vulnerabilities

Devices on outdated operating systems, or using unpatched firmware exposes the network to known vulnerabilities. Open ports can lead to port scanning and DDoS (Denial of Service) attacks, as well as a deficit in rate-limiting. Not using proper authentication methods creates entry points vulnerable to brute force and credential stuffing attacks. Additionally, flaws in existing software means they can be exploited, further leaving the system open and vulnerable.

Credential Management

Using default or weak passwords (e.g. "admin/admin") on network devices leaves the system open to unauthorised access. Password reuse across multiple accounts increases the risk of credential compromise.

Inadequate Endpoint and Network Security Controls

If endpoints are used, i.e. desktops, laptops and servers, without active antivirus or anti-malware solutions they are vulnerable to ransomware, trojans and other malicious infections. Firewalls which have overly permissive rules and are missing critical access controls expose data to external threats. Additionally, unsecure file sharing networks allow unauthorised access to sensitive data.

Human Factors

Social Engineering

Phishing attacks which leverage fake login pages, malicious email attachments, and exploiting links are common methods which compromise user credentials and allow attackers to gain network access. People can be manipulated into revealing user details, and can overshare on social media accounts. By playing to a person's feelings, attackers can easily gain access if staff are not prepared.

Physical Security

Physical security can be compromised by unauthorised physical access through many methods. For example, tailgating or lax entry controls allow attackers to interact and interfere directly with networking equipment and or endpoints. The use of personal devices can also lead to a compromise of the network, especially if they are not properly configured or secured. They can introduce malware or bypass corporate security controls.

Organisational Weaknesses

Security Governance and Resource Limitations

A lack of documented security policies which relate to device usage, password requirements and data handling results in inconsistent and ununified security practices. This lack of uniformity, alongside a lack of regular security audits leaves potential vulnerabilities open and the overall network unprotected. Additionally, limited staff or absence of dedicated cybersecurity personnel increases the time to respond to threats, as well as patching vulnerabilities.

Lack of Incident Management

When backups are not performed regularly, it heightens the risk of permanent data loss from ransomware attacks or system failures. Undertrained staff, as well as a lack of a proper incident response

plan leads to delayed and ineffective breach containment, exacerbating damage. Also, absence of cybersecurity insurance leaves the enterprise financially vulnerable, as well as ethical and legal consequences.

Third-Party Security Risks

Granting administrative power or broad access rights to all users on the network increases the potential damage from insider threat and compromised accounts. Third party vendors or contractors with full network access may introduce vulnerabilities, especially if their cybersecurity status is inadequate or non-aligned with organisational standards.

Risk matrix

		Likelihood				
		Rare 1	Unlikely 2	Possible 3	Likely 4	High Likely 5
Impact	Negligible 1	1	2	3	4	5
	Minor 2	2	4	6	8	10
	Moderate 3	3	6	9	12	15
	Severe 4	4	8	12	16	20
	Very Severe 5	5	10	15	20	25
Low, Medium, Medium High, High, Very High						

Risk	Impact	Likelihood	Score	Priority
Outdated operating systems and firmware	5	4	20	Very High
Open ports and lack of rate-limiting	4	4	16	High
Weak/default passwords	5	4	20	Very High
Password reuse across multiple accounts	4	4	16	High

Lack of antivirus/endpoint protection	5	4	20	Very High
Misconfigured firewalls	4	3	12	Medium High
Unsecured file sharing	4	3	12	Medium High
Phishing attacks	5	5	25	Very High
Social engineering	4	4	16	High
Tailgating and poor physical access control	3	3	9	Medium
Insecure personal devices containing malware	4	3	12	Medium High
Lack of formal security policies and implementation	5	4	20	Very High
Absence of regular audits and understaffed IT team	4	4	16	High
No regular backups	5	3	15	High
No incident response plan	5	3	15	High
Lack of cybersecurity insurance	4	3	12	Medium High
Overly permissive access rights	4	4	16	High
Third-party vendor risks	4	3	12	Medium High
Insider threats	5	2	10	Medium High

IoT Device Vulnerabilities	4	3	12	Medium High
Zero-day exploits	5	2	10	Medium
Compromised software updates	5	3	15	High
Inadequate Data Encryption	5	3	15	High
Insufficient Logging and Monitoring	4	4	16	High
Ransomware	5	5	25	Very High

Mitigation Plan

Technical Measures

Updates should be regularly maintained on operating systems, firmware and third-party applications. This reduces the risk of known vulnerabilities that attackers can exploit. A structured patch management program should be implemented. Network hardening should be conducted by disabling used ports and configuring strong firewalls using the principle of least privilege (PoLP). The principle of least privilege revolves around users only having a level of access suitable to what is necessary for their needs. This reduces the risk of port scanning and insider attacks. On this local business, unused ports on the switches have been turned off, and wireless access points have been configured to only allow specific MAC addresses to connect. If an unauthorised MAC address connects, the wireless access point will turn off until a restart of the point which has to be done manually. If a new user wants to connect the network, the IT Wireless Router has two guest networks set up, giving basic access to the internet, keeping the rest of the network private. Strong credential management should be enforced through complex passwords, which are regularly changed. Multifactor authentication should be used, significantly decreasing the risk of brute force and credential stuffing attacks. Endpoints should also be protected using antiviruses and monitoring tools, ensuring malware is detected and dealt with early. This reduces the impact of ransomware and trojan attacks. Sensitive data should be encrypted, minimising the impact of breaches, ensuring critical stolen data cannot be read or exploited.

Employee Training and Awareness Programs

Social engineering is one of the most common cyber attacks. They manipulate staff into providing access to the network, presenting a very critical risk to the local business. To reduce the likelihood of social engineering and human error, all staff will undergo mandatory cybersecurity awareness training which will teach them about how to react to phishing and social engineering attacks, as well as safe password practices and how to appropriately use personal devices. Regular phishing simulations will ensure vigilance, and help employees recognise real-world threats. For physical security, staff will be briefed on tailgating as well as reporting suspicious activities/individuals, reducing the risk of physical access to networking equipment and endpoints. Staff should not overshare business details on social media to prevent leakage of private information. Additionally, each department will feature dedicated IT support/cybersecurity personnel to ensure rapid response and aid to any issues which arise.

Policy and Procedural Changes

The local business should establish formal cybersecurity policies, involving acceptable use of the network, password requirements, data handling standards, and personal device (BYOD) restrictions. This is critical to both reducing risk likelihood and limiting impacts when incidents occur. Regular audits and vulnerability assessments conducted by cybersecurity personnel will help identify and address gaps before they can be exploited. The local business must maintain a robust backup system, including daily incremental and weekly full backups which are stored offsite. An incident response plan, which assigns responsibilities and defines escalation procedures, should be created to ensure that when a breach occurs everyone knows their part, ensuring the breach is contained quickly. Finally, cybersecurity insurance will provide financial security and resilience, reducing the long-term impact of an otherwise catastrophic security incident.

Vendor and Third-Party Risk Management

External entities, such as contractors, often have privileged access to business information and networks, posing that as a potential attack vector. To mitigate the risk, the local business will require all third parties to comply with equivalent security standards, with contracts including clauses that mandate incident notification and outline liability in a breach. If possible, external access should be restricted to the minimum required systems, following PoLP. with them being monitored through logging and alerting. Regular vendor risk assessments will be conducted, alongside security questionnaires, further reducing the likelihood of supply chain attacks. By managing vendor and third party risks, the local businesses decrease the chance of an external breach, as well as reducing the potential impact.

Bibliography

A comprehensive guide to cyber security protocols and best practices.

(2024, September 12). DataGuard. Retrieved August 27, 2025, from <https://www.dataguard.com/blog/cyber-security-protocols-and-best-practices/>

Cybersecurity guidelines | Cyber.gov.au. (n.d.). Australian Cyber Security Centre. Retrieved August 27, 2025, from <https://www.cyber.gov.au/resources-business-and-government/essential-cybersecurity/ism/cybersecurity-guidelines>

Cybersecurity risk assessment. (n.d.). ManageEngine. Retrieved August 27, 2025, from <https://www.manageengine.com/data-security/what-is/cybersecurity-risk-assessment.html>

Deposit Photos. (n.d.). Apartment Architectural Plan. Deposit Photos. Retrieved August 2, 2025, from <https://depositphotos.com/vector/apartment-architectural-plan-top-view-floor-plan-vector-blueprint-project-650969468.html>

Genau 4 NOW. (2019, April 8). Cisco Packet Tracer Tutorial # 3 | Internet access with the network we built. Youtube. Retrieved August 2, 2025, from <https://www.youtube.com/watch?v=FlxF4tiJ1zU>

Grumpy_Grampy / JasonEJohnson31. (2022, February 17). Cisco Packet Tracer 📶 Add IoT Devices to a Basic Wireless Network. Youtube. Retrieved August 2, 2025, from <https://www.youtube.com/watch?v=l16reIl3tv4>

Jain, S. (2025, July 23). Advantages and Disadvantages of Bus Topology. GeeksforGeeks. Retrieved August 2, 2025, from <https://www.geeksforgeeks.org/computer-networks/advantages-and-disadvantages-of-bus-topology/>

Jain, S. (2025, July 23). *Advantages and Disadvantages of Tree Topology*. GeeksforGeeks. Retrieved August 2, 2025, from <https://www.geeksforgeeks.org/computer-networks/advantages-and-disadvantages-of-tree-topology/>

Jain, S. (2025, July 23). *Implementation of Tree Topology in Cisco*. GeeksforGeeks. Retrieved August 2, 2025, from <https://www.geeksforgeeks.org/computer-networks/implementation-of-tree-topology-in-cisco/>

Ptnetcad. (n.d.). *Official Packet Tracer Tutorials*. Ptnetcad. Retrieved August 2, 2025, from <https://tutorials.ptnetacad.net/#stub>

Strategies to mitigate cyber security incidents | Cyber.gov.au. (n.d.). Australian Cyber Security Centre. Retrieved August 27, 2025, from <https://www.cyber.gov.au/resources-business-and-government/essential-cyber-security/strategies-mitigate-cyber-security-incidents>